1) Consider the three-link planar manipulator shown in **Figure 1**. Derive the forward kinematic equations using the DH convention.

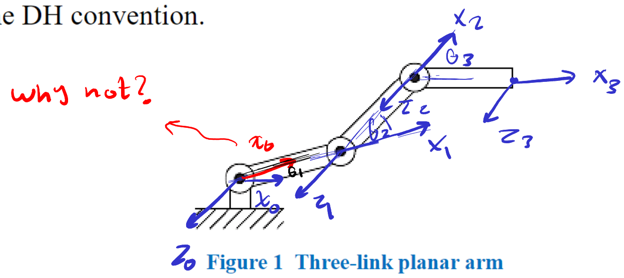


Figure 1 Three-link planar arm

	θ	d	a	α
1	θ_1^*	0	L_1	0
2	θ_2^*	0	L_2	0
3	θ_3^*	0	L_3	0

2) Consider the two-link Cartesian manipulator shown in **Figure 2**. Derive the forward kinematic equations using the DH convention.

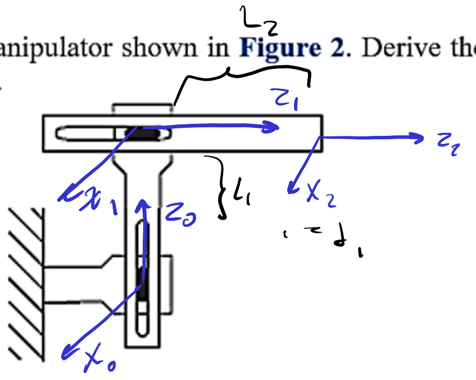


Figure 2 Two-link Cartesian robot

	θ	d	a	α
1	0	d_1^*	0	-90°
2	0	d_2^*	0	0

3) Consider the two-link manipulator of **Figure 3**, which has joint 1 revolute and joint 2 prismatic. Derive the forward kinematic equations using the DH convention.

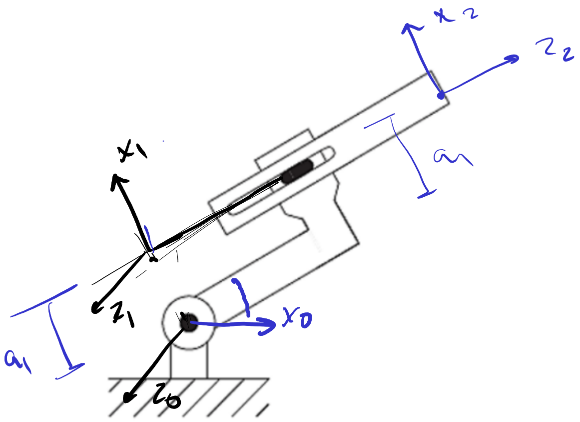


Figure 3 Two-link planar arm

$\theta_1^* = \theta_1 + 90$

	θ	d	a	α
1	θ_1^*	0	a_1	0
2	0	d_2^*	0	0

5) Consider the three-link articulated robot of **Figure 5**. Derive the forward kinematic equations using the DH convention.

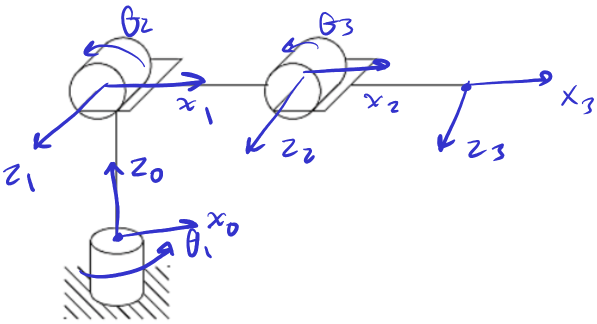


Figure 5 Three-link articulated robot

	θ	d	a	α
1	θ_1^*	d_1	0	90^0
2	θ_2^*	0	a_2	0
3	θ_3^*	0	a_3	0

4) Consider the three-link planar manipulator of **Figure 4**. Derive the forward kinematic equations using the DH convention.

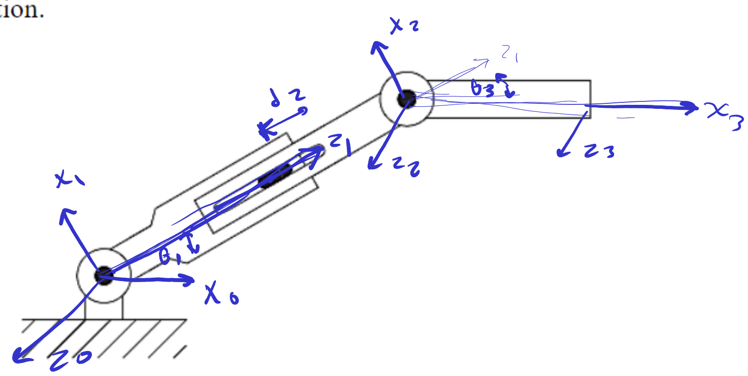


Figure 4 Three-link planar arm with prismatic joint

	θ	d	a	α
1	θ_1^*	0	0	90
2	0	d_2^*	0	-90
3	θ_3^*	0	a_3	0

$\theta_1^* = \theta_1 + 90$

$\theta_3^* = \theta_3 + 90$

6) Consider the three-link Cartesian manipulator of **Figure 6**. Derive the forward kinematic equations using the DH convention.

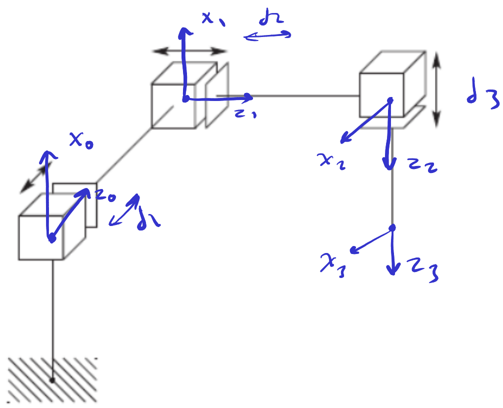
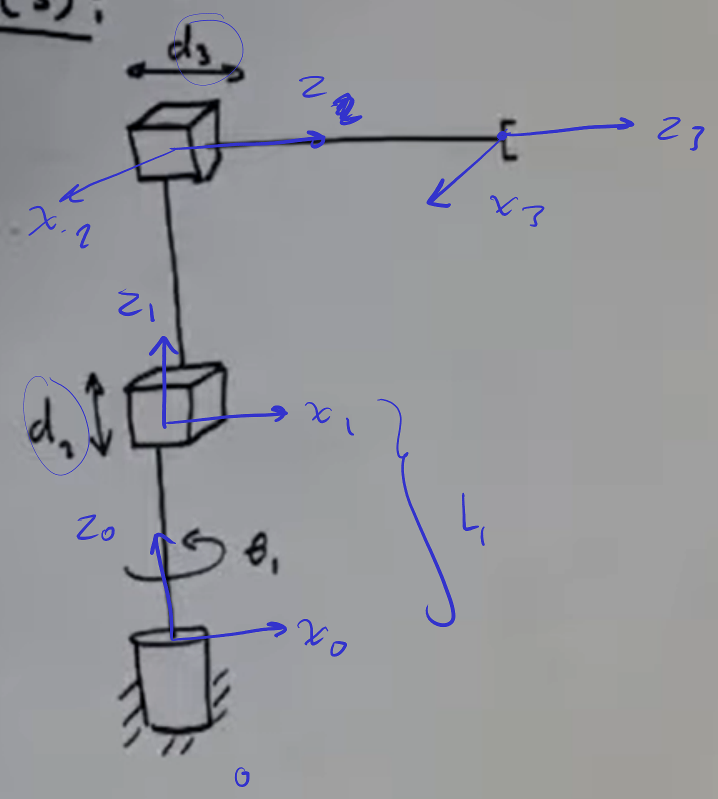


Figure 6 Three-link Cartesian robot.

	θ	d	a	α
1	0	d_1^*	0	-90
2	90	d_2^*	0	-90
3	0	d_3^*	0	0

	θ	d	a	α
1				
2				
3				

Example (3):



i	α_i	a_i	d_i	θ_i
1	0	0	L_1	θ_1^*
2	-90	0	d_2^*	-90
3	0	0	d_3^*	0